2: axiom:

$\exists c$, unique, s.t., $a(S) \le c \le a(T)$
$\forall$ step regions S, T, satisfying
$S \subseteq Q \subseteq T$.

$\Rightarrow Q$ measurable $\Rightarrow a(Q) = c$
(monotone property)

(here Q a set that can be enclosed by some S, T, i.e., $\exists$ step regions S, T, $S \subseteq Q \subseteq T$).

2.3 sup, inf $\exists$, and sup = inf,
i.e., $\sup\{a(S) \mid \text{step region } S \subseteq Q\}$
$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

$\Rightarrow Q$ measurable (use 1)
$\Rightarrow a(Q) = \sup = \inf$ (use monotone property)

(here set Q is enclosed by some step regions)

(here Q is a set that can be enclosed by some step regions).

proof:

1) $\Rightarrow$

① $\because Q$ enclosed by some step regions,
$\therefore \{a(S) \mid S \subseteq Q\}$, $\{a(T) \mid T \supseteq Q\}$ nonempty.
$\because a(S) \le c \le a(T)$, $\forall S \subseteq Q \subseteq T$
$\therefore$ sup, inf $\exists$, and sup $\le$ inf

② $\forall c' \in [\sup, \inf]$,
$a(S) \le c' \le a(T)$, $\forall S, T, S \subseteq Q \subseteq T$
$\because c'$ must be unique,
$\therefore$ sup = inf.

2) $\Leftarrow$

① $\because$ sup = inf $\therefore \exists c = \sup = \inf$
$\forall$ step region $S \subseteq Q$, $a(S) \le c$
$\forall$ step region $T \supseteq Q$, $a(T) \ge c$

② if $\exists c'$, s.t., $a(S) \le c' \le a(T)$ $\forall S \subseteq Q \subseteq T$
then, $\sup \le c' \le \inf$
$\because$ sup = inf $\therefore c' = c$
$\therefore c$ is unique.

$\boxed{a(\text{graph of } f) = 0}$

0.1 rectangle: any set congruent to the set
$\{(x,y) \mid 0 \le x \le h, 0 \le y \le k\}$
where $h \ge 0$, $k \ge 0$

0.2 step region: the union of a finite collection of adjacent rectangles with their bases resting on the x-axis.

0.3 ① $\forall$ step region $S \Rightarrow \exists$ naturally step function $s \ge 0$,
s.t., $a(S) = \int_a^b s(x)\,dx$

② $\forall$ step function $s \ge 0$, $\exists$ naturally step region S, s.t., $a(S) = \int_a^b s(x)\,dx$.

1. $\exists c$, unique, s.t., $a(S) \le c \le a(T)$
$\forall$ step regions S, T, $S \subseteq Q \subseteq T$.
$\Leftrightarrow \sup\{a(S) \mid \text{step region } S \subseteq Q\}$
$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

5. step function $s \geq 0$, defined on $[a,b]$,
Q ordinate set of s,
$\Rightarrow$ Q measurable, $a(Q) = \int_a^b s(x)dx$

6. $f \geq 0$, integrable on $[a,b]$,
① ordinate set of f over $[a,b]$
(here $Q = \{(x,y) | a \leq x \leq b, 0 \leq y \leq f(x)\}$)
$\Rightarrow$ ① $\int f = \sup\{a(S) | S \subseteq Q, S \text{ step region}\}$
$= \inf\{a(T) \, T \supseteq Q, T \text{ step region}\}$
② Q measurable, $a(Q) = \int_a^b f(x)dx$

prof: ① f integrable on $[a,b]$ $\Rightarrow$ f defined and bounded on $[a,b]$
$f \geq 0 \Rightarrow Q \, \exists$.
$\therefore$ Q is bounded by some step regions.
② $\forall$ step region $S \subseteq Q$, $\exists$ step function $0 \leq s \leq f$,
s.t., $a(S) = \int s$ (from 0.3)
$\therefore \{a(S) | S \subseteq Q\} \subseteq \{\int s | 0 \leq s \leq f\}$
$\forall$ step fun $0 \leq s \leq f$, $\exists$ step region $S \subseteq Q$,
s.t., $a(S) = \int s$ (from 0.3)
$\therefore \{\int s | 0 \leq s \leq f\} \subseteq \{a(S) | S \subseteq Q\}$
$\therefore \{\int s | 0 \leq s \leq f\} = \{a(S) | S \subseteq Q\}$

② if $\exists I'$, s.t., $\int s \leq I' \leq \int t$, $\forall s,t$, $s \leq f \leq t$
$\Rightarrow I' \in [\sup, \inf]$.
$\because \sup = \inf$ $\therefore I' = I$, $\therefore I$ unique

4. definition
$\exists I$, unique, s.t., $\int_a^b s(x)dx \leq I \leq \int_a^b t(x)dx$
$\forall$ step functions s, t, satisfying
$s(x) \leq f(x) \leq t(x)$
$\Leftrightarrow f$ integrable (definition)
$\Rightarrow \int_a^b f(x)dx = I$ (use comparison theorem)
(here f defined and bounded on $[a,b]$)

4.3 $\sup = \inf$, i.e.,
$\sup\{\int_a^b s(x)dx \,|\, \text{step function } s \leq f\}$
$= \inf\{\int_a^b t(x)dx \,|\, \text{step function } t \geq f\}$
$\Leftrightarrow f$ integrable
$\Rightarrow \int f = \sup = \inf$
(here f defined and bounded on $[a,b]$)

3. $\exists I$, unique, s.t.,
$\int_a^b s(x)dx \leq I \leq \int_a^b t(x)dx$
$\forall$ step functions s, t, satisfying
$s(x) \leq f(x) \leq t(x)$
$\Leftrightarrow$ sup, inf $\exists$ and $\sup = \inf$, i.e.,
$\sup\{\int_a^b s(x)dx \,|\, \text{step function } s \leq f\}$
$= \inf\{\int_a^b t(x)dx \,|\, \text{step function } t \geq f\}$
(here f defined and bounded on $[a,b]$)

prof: $\Rightarrow$ ① $\because f$ bounded
$\therefore \{\int s | s \leq f\}, \{\int t | t \geq f\}$ nonempty
$\because \int s \leq \int t$ if $s \leq f \leq t$ $\therefore$ sup, inf $\exists$
and $\sup \leq \inf$
② $\forall c' \in [\sup, \inf]$, $\int s \leq c' \leq \int t$, $\forall s \leq f \leq t$
$\because c'$ unique $\therefore \sup = \inf$
$\Leftarrow$ ① $\because \sup = \inf$, $\therefore \exists I = \sup = \inf$
$\int s \leq I \leq \int t$, $\forall s,t$, $s \leq f \leq t$

② Q' measurable, $a(Q') = \int f$

③ $a(\text{graph of } f) = 0$

proof: $\because \int f = \sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

(from 6)

$\therefore \int f = \sup\{a(S') \mid S' \subseteq Q'\} = \inf\{a(T') \mid T' \supseteq Q'\}$

(from 7)

$\therefore Q'$ measurable, $a(Q') = \int f$ (from 2,3)

$\therefore a(\text{graph of } f) = a(Q) - a(Q') = 0$

(from difference property of area axiom)

9. $f \geq 0$, integrable on $[a,b]$, Q', Q from 8

$\Rightarrow$ ① Q' measurable, $a(Q') = \int f$. ② $a(\text{graph of } f) = 0$

proof: let $g(x) = f(x) + 1$ $\therefore g(x) > 0$, g integrable on $[a,b]$

(basic property of integral)

$\therefore a(\text{graph of } g) = 0$ (from 8)

$\therefore a(\text{graph of } f) = 0$

(invariance under congruence of area axiom)

$\therefore Q'$ measurable, $a(Q') = a(Q) - a(\text{graph of } f)$

$= \int f$ ▨.

10. f integrable on $[a,b] \Rightarrow a(\text{graph of } f) = 0$

proof: ① f integrable $\Rightarrow f$ bounded, let $m \leq f(x) \leq M$

② let $g(x) = f(x) - m$ $\therefore g \geq 0$, g integrable on $[a,b]$

$\therefore a(\text{graph of } g) = 0$ $\therefore a(\text{graph of } f) = 0$ ▨

$\forall \varepsilon > 0, \exists S \subseteq Q$, $a(S) > \sup - \varepsilon$.

$\therefore \exists S' \subseteq Q'$, $a(S') > \sup - 2\varepsilon$.

$\therefore \sup = \sup'$.

detail: let S: height s_i if $x \in (x_{i-1}, x_i)$

S': height $s_i' = \max(s_i - \delta, 0)$, if $x \in (x_{i-1}, x_i)$

$i = 1, \cdots, n$.

$P = \{x_0, x_1, \cdots, x_n\}$ be partition of $[a,b]$

$\because a(S) - a(S') \leq \delta(b-a)$

$\therefore$ choose $\delta < \frac{\varepsilon}{b-a}$, we have $a(S) - a(S') < \varepsilon$

② $\because Q' \subseteq Q$, $\therefore \{a(T) \mid T \supseteq Q\} \subseteq \{a(T') \mid T' \supseteq Q'\}$

$\forall T' \supseteq Q'$, let T': height t_i if $x \in (x_{i-1}, x_i)$

$\because T' \supseteq Q' = \{(x,y) \mid a \leq x \leq b, 0 \leq y < f(x)\}$

$\therefore t_i \geq f(x)$, if $x \in (x_{i-1}, x_i)$

$\therefore T' \supseteq Q$ $\therefore \{a(T) \mid T \supseteq Q\} \supseteq \{a(T') \mid T' \supseteq Q'\}$

$\therefore \{a(T) \mid T \supseteq Q\} = \{a(T') \mid T' \supseteq Q'\}$

8. $f > 0$, integrable on $[a,b]$, ▨

$Q' = Q - \text{graph of } f$,

Q ordinate set of f over $[a,b]$.

here $Q' = \{(x,y) \mid a \leq x \leq b, 0 \leq y < f(x)\}$

$\Rightarrow$ ① $\int f = \sup\{a(S') \mid \text{step region } S' \subseteq Q'\}$

$= \inf\{a(T') \mid \text{step region } T' \supseteq Q'\}$

③ similarly, $\{\int t \mid t \geq f\} = \{a(T) \mid T \supseteq Q\}$

④ $\because f$ integrable $\Rightarrow \int f = \sup\{\int s \mid 0 \leq s \leq f\}$

$= \inf\{\int t \mid t \geq f\}$

(from 4,3)

$\therefore \int f = \sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \inf\{a(T) \mid \text{step region } T \supseteq Q\}$

$\therefore Q$ measurable, $a(Q) = \int f$ (from 2,3)

▨

7. $f > 0$, integrable on $[a,b]$,

$Q' = Q - \text{graph of } f$,

Q is ordinate set of f over $[a,b]$

$\Rightarrow$ ① $\sup\{a(S) \mid \text{step region } S \subseteq Q\}$

$= \sup\{a(S') \mid \text{step region } S' \subseteq Q'\}$

② $\{a(T) \mid \text{step region } T \supseteq Q\}$

$= \{a(T') \mid \text{step region } T' \supseteq Q'\}$

proof: ① $\because Q' \subseteq Q$

$\therefore \sup = \sup\{a(S) \mid S \subseteq Q\} \geq \sup\{a(S') \mid S' \subseteq Q'\}$

$= \sup'$

(sup' must exist)